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Key Personnel Changes/Previous PI:	November 2012 report: Dr. Sucharita Dutta, a proteomics expert, has been added to the team.		
COI Name (Institution):	Semmes, Oliver (Eastern Virginia Medical School) Dutta, Sucharita (Eastern Virginia Medical School)		
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Task Description:	One of NASA's biggest concerns about the long-term health of astronauts who go on deep space missions is the impact that Hze radiation will have on brain function (neurocognition). Data from our laboratory and others suggests that there is significant impairment of certain neurocognitive tasks following exposure to low Hze doses. The goal of this application is to use proteomic profiling/imaging to establish the mechanistic basis of Hze-induced cognitive impairment. Our underlying hypothesis is that Hze-induced spatial memory impairments (HISMI) or Attentional Set Shifting Impairments (HIASSI) are due to Hze-induced changes in the proteome of the brain (neuroproteome). Our sub-hypothesis is that the use of a novel technique, called MALDI-MSI, will identify specific regions of the brain where the neuroproteome differs in rats that have developed HISMI and HIASSI. We further hypothesize that proteomic profiling of serum and cerebrospinal fluid (CSF) will identify proteins that can be used as biomarkers to monitor the emergence of HISMI and HIASSI. We also hypothesize that different mechanisms of HISMI and HIASSI occur following exposure to Hze radiations with different track structures. Our studies will thus address Central Nervous System (CNS) Gap 2, and we shall specifically focus of the following aims: Aim 1. Identify changes in the neuroproteome that are associated with HISMI and HIASSI. Aim 2. Identify proteins that are present in the serum and CSF that are biomarkers of an individual rat¿s susceptibility to HISMI/HIASSI. Aim 3. Determine the Linear Energy Transfer (LET)-dependency for HISMI and HIASSI, and whether the threshold dose is modified by prior exposure to protons, and if the underlying mechanisms of HISMI and HIASSI are constant irrespective of the LET of the Hze particle. In Aim 1, two groups of rats (5-week and 9 month old males) will be irradiated with 20 cGy Hze (1 GeV 56Fe) irradiation, and HISMI and HIASSI assessed at 3 months post irradiation. The rats will then be classified as normal or as having HISMI or HIASSI, then sacrificed, and their brains recovered. We shall use MALDI-IMS to analyze changes in the neuroproteome at the regional level. Subsequently, we shall identify specific proteins whose expression changes when HISMI or HIASSI have emerged, and will use immunohistochemcial analysis to determine the specific brain tissue (e.g., the dentate gyrus or CA2) affected. These studies will give considerable insight into the underlying cause for Hze-induced neurocognitive failure. In Aim 2, biological fluids (CSF and serum) from the rats used in Aim 1 will be recovered, and then subjected to MALDI-TOF/TOF proteomic profiling to identify unique proteins that are prognostic markers of an individual's rat's susceptibility to develop HIMSI and HIASSI. These proteins could be potentially used for post-mission surveillance of astronauts to monitor the emergence of Hze-induced neurocognitive impairments. In Aim 3, we shall repeat the studies outlined in Aim 1, but determine the impact that exposure to other Hze particles (300 MeV 37Cl, 1 Gev 48Ti , 400 MeV 48Ti) have on the induction of HISMI and HIASSI, and determine whether the mechanism of HIASSI and HISMI is the same following exposure to these Hze radiations as was observed in Aim 1. These studies will be confined to the aged rat population, as these resemble the biological age of the current astronaut cadre. We shall also determine if prior irradiation with 1 Gy of protons modifies the threshold dose for HISMI and HIASSI and the spectrum of proteomic changes that occur when HIASSI and HISMI emerge. Due to logistical constraints these combined proton/Hze exposure studies will be confined to 400 MeV 48Ti and 1 GeV 56Fe. Collectively, these studies will provide considerable insight into the underlying mechanism of Hze-induced neurocognitive failure and potentially provide a post mission screen for HISMI and HIASSI in astronauts returning from deep space missions.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	Current models predict that the astronauts on deep space missions will be exposed to ~25 cGy of Galactic cosmic radiation (GCR). The long-term consequence of exposure to such doses is largely unknown, but recent data from our laboratory and that of Dr. Rabin suggest that low (<20 cGy) doses of HZE particles lead to significant impairments of various neurocognitive functions. We have shown that exposure of young rats to 20 cGy of 1 GeV/µ 56Fe leads to a significant impairment in spatial memory (Britten et al., 2012) and attentional set shifting performance (Lonart et al., 2012). Data from Dr. Rabin suggest that the LET dependency for neurocognitive impairment may differ substantially from that seen for cell killing. We have thus started to determine the LET-dependency for impairment of spatial memory performance and attentional set shifting in rats that have a comparable biological age to the astronaut population. Data has now been obtained from at least three batches of rats exposed to 1 GeV/µ 56Fe, 1.1 GeV/µ 48Ti and 600 MeV/µ 28Si (respective LETs: 150, 106, and 51 keV/um). We have also started to investigate the mechanistic basis for the impairment of these various neurocognitive processes using an unbiased proteomic analysis. Our initial efforts have focused on determining the changes in the proteome of the hippocampus in rats that have good or poor spatial memory performance after exposure to 20 cGy of 1 GeV/µ 56Fe and 1.1 GeV/µ 48Ti particles. MATERIALS & METHODS. Retired breeder (6-9 months old) male Wistar rats were exposed to either 1 GeV/µ 56Fe, 1.1 GeV/µ 48Ti, or 600 MeV/µ 28Si particles at BNL. At 3 months post exposure the ability of rats to conduct attentional set shifting and spatial memory was established. After completion of the behavioral tasks the animals were euthanized and the brains recovered for subsequent protoeomic analysis on the Q-Exactive platform. RESULTS. Spatial memory impairment. Data from at least three cohorts of aged rats, suggests that spatial memory impairment occurs following exposure to 10 cGy (but not after 5 cGy) 1 GeV/u 56Fe. However, there is a marked inter-individual susceptibility to develop spatial memory impairment, with ~33% of rats exposed to 20 cGy exhibiting spatial memory performance comparable to that observed in the sham-irradiated rats. Exposure of aged rats to 15 cGy of 1.1 GeV/u 48Ti ions resulted in an impairment of spatial memory performance, but in contrast to what was observed with 56Fe particles, no impairment was seen following exposure to 10 cGy 1.1 GeV/u 48Ti. Our preliminary data (2 batches, N<15 per dose point) suggest that Aged rats have impaired spatial memory following exposure to 10 cGy 600 MeV/µ 28Si particles. It is clear that exposure to mission-pertinent doses of the HZE species studied (LETs ranging from ~50-150 KeV/µm) results in measurable spatial memory impairment (10 cGy: 600 MeV/µ 28Si and 1 GeV/µ 56Fe; 15 cGy: 1.1 GeV/µ 48Ti). However, there is no obvious variation in the potency of these HZE particles to induce spatial memory impairment. Attentional Set shifting impairment. We recently published (Lonart et al., 2012) data from young rats that had received cranial irradiation with 1 GeV/µ 56Fe, where we reported that exposure to 20 cGy 1 GeV/µ 56Fe resulted in the majority of rats being incapable of completing all stages of the Attentional Set Shifting test, with many rats failing at the Reversal 1 stage. Whole body exposures of young rats resulted in a similar pattern of failures. Sixty percent of the sham treated rats completed all stages of the testing whereas only 20% of the irradiated rats could complete all stages of the task. However, these rats have a biological age comparable to that of teenagers and our primary mandate is now to determine if rats with a comparable age to the Astronauts were equally sensitive to the effects of these low doses of 1 GeV/µ 56Fe, 1.1 GeV/µ 48Ti, and 600 MeV/µ 28Si particles. The studies with the Aged rats have been complicated by a marked decline in the ability, or the motivation of sham-irradiated Aged (SIA) rats to conduct the first paradigm of the testing, i.e. Simple Discrimination (regulated by the medial prefrontal cortex). Four batches of aged rats that have received whole-body irradiation with 10, 15, or 20 cGy 1 GeV/µ 56Fe, have now been screened, with a minimum of 15 rats per dose point. Rats exposed to 10 cGy 1 GeV/µ 56Fe were indistinguishable from the SIA rats in their ability to conduct the various paradigms of the test. However, rats exposed to 15 cGy 1 GeV/µ 56Fe were ~2-fold less capable than the SIA rats of completing the Simple and Complex Discrimination paradigms (tasks regulated by the medial prefrontal cortex and perirhinal cortical region respectively). Rats exposed to 20 cGy 1 GeV/µ 56Fe were 5-fold less capable than the SIA rats of completing the habituation stage of the test. The majority of the rats that failed to habituate started to dig for the reward but quickly gave up, suggesting they understood the concept of the test, but were either lethargic, mentally fatigued, or unmotivated to dig for the reward. Those rats that did complete the habituation task completed the remaining paradigms with an identical success rate as did the sham-irradiated animals. Four batches of aged rats irradiated with 10, 15, or 20 cGy 1.1 GeV/u 48Ti ions, with a minimum of 15 rats per dose point, have now been screened. Rats exposed to 10 and 20 cGy were 3-fold less capable than the SIA rats of completing the habitation stage of the test. Rats exposed to all doses were ~ 2-fold less capable than the SIA rats of completing the Reversal 1 stage (regulated by the basal forebrain), while rats exposed to 10 and 15 cGy also had a decreased ability to complete the Intra-dimensional shifting paradigm (regulated by the anterior and posterior cingulated cortex). The first two batches of rats exposed to 600 MeV/µ 28Si ions have now been screened for attentional set shifting performance. Exposure to 10 cGy resulted in 7-fold more rats failing the habituation stage of the testing than was observed with the SIA rats. Exposure to 15 cGy resulted in a 2.5 fold higher failure rate at both the Simple and Discrimination paradigms. Rather paradoxically, rats exposed to 20 cGy resembled the SIA rats in their performance in all tasks. These data suggest that exposure to mission-relevant HZE doses leads to a decreased ability of rats to habituate to the task, which appears to be related to mental fatigue or lack of motivation to dig for the reward, rather than a lack of comprehension of the task. For those rats that did habituate to the test, the Discrimination tasks (and the reversal of those associative clues) are most affected by mission-relevant doses of HZE (1 GeV/u 56Fe, 1.1 GeV/u 48Ti, and 600 MeV/u 28Si) particles. However, due to the current practice of ceasing testing of rats once they fail a task, later tasks may be impaired. Hippocampal proteomic changes. For this work we partnered with The Biomarkers Research Initiatives in Mass Spectrometry (BRIMS) Center, a Thermo Fisher Scientific (http://www.thermoscientific.com) center of excellence that develops and promotes comprehensive, integrated, and robust mass spectrometry (MS)-based workflows that links early-stage discovery to next-stage quantitative verification of protein, peptide, and small molecule biomarkers. Using this workflow we have been able to obtain quantitative data on ~900 proteins in most hippocampal sample studied. This wealth of data has allows us to determine some of the changes in the hippocampal proteome of the rats exposed to mission relevant doses of 1 GeV/µ 56Fe particles, and the specific changes observed in 20/Impaired and 20/Functional rats has provided some insight into the functionality of multiple pathways in these rats. While quantitative data has been derived on ~1000 proteins, it must be remembered that the proteomic profiling technique used in this study, while being one of the most sensitive currently available, has not detected low abundance proteins that are likely to be the key rate limiting steps in many pathways. Nonetheless, there is clear evidence that hippocampi exposed to the HZE radiation have persistent oxidative stress, altered adenosine metabolism and may have on-going GluR1-mediated neuronal death. However, the ultimate phenotypic outcome (functional vs. impaired spatial memory performance) appears to be determined by either the up-regulation of pathways that serve to ameliorate the aforementioned deficits, by altering brain plasticity (20/Functional rats), or failure to invoke, or even lose key proteins in those pathways (20/Impaired). This work is in the final stages of being prepared for submission for publication to a neuroscience journal. SUMMARY. Collectively, these data suggest that mission-relevant doses (10-15 cGy) of HZE particles with LETs of 51-150 KeV/µm induce significant neurocognitive defects (spatial memory performance and attentional set shifting) in rats of comparable biological age to the astronaut population. However, particles with different LETs may affect different aspects of executive function and may possibly induced mental fatigue or depression-like symptoms. We have identified specific changes in the hippocampal proteome that are associated with either good or bad spatial memory performance in HZE-irradiated rats, that may provide some insight into the mechanisms that govern individual susceptibility to HZE-induced cognitive impairment.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Britten RA, Davis LK, Jewell J, Lonart GL. "Executive function is significantly impaired following exposure to low (2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Abstracts for Journals and Proceedings	Britten RA, Dutta S, Davis L, Jewell J, Miller V, Semmes OJ. "Changes in the hippocampal proteome associated with the induction of Spatial memory impairment by mission-relevant HZE doses." Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. , May-2013
Abstracts for Journals and Proceedings	Britten RA, Dutta S, Davis L, Jewell J, Miller V, Semmes OJ. "Changes in the hippocampal proteome associated with the induction of Spatial memory impairment by mission-relevant HZE doses." 59th Annual Meeting of the Radiation Research Society, New Orleans, LA, September 14-18, 2013. 59th Annual Meeting of the Radiation Research Society, New Orleans, LA, September 14-18, 2013. Presentation Number: S1403. Abstract available at: http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=f5102ea8-6533-47c9-8a2b-808c4b953413&cKey=563aedb9-c6fa-4cfc-9095-d68fc9a8193b&mKey=01c994ce-9545-4cc7-8580-d74360fa8373 ; accessed 1/27/14. , Sep-2013